

Classical Mechanics 1, Autumn 2025 CMI

Assignment 8

Due by 6pm, 14 Nov, 2025

Kepler problem

1. **⟨3 + 4 + 4⟩ CM and relative variables.** The center of mass and relative vectors in the problem of 2 point particles of masses m_1, m_2 and positions $\mathbf{r}_1, \mathbf{r}_2$ are defined as $\mathbf{R} = (m_1\mathbf{r}_1 + m_2\mathbf{r}_2)/M$ and $\mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1$ where $M = m_1 + m_2$. Let $\mathbf{p}_1$ and $\mathbf{p}_2$ be the momenta of the two particles. Let us define $\mu_{1,2} = m_{1,2}/M$ and the reduced mass $m = m_1m_2/(m_1 + m_2)$. (a) Draw a figure with a suitable origin (not at the center of mass) indicating the vectors $\mathbf{R}, \mathbf{r}, \mathbf{r}_1, \mathbf{r}_2$. (b) Show that

$$\mathbf{r}_1 = \mathbf{R} - \mu_2\mathbf{r}, \quad \mathbf{r}_2 = \mathbf{R} + \mu_1\mathbf{r}. \quad (1)$$

Let the CM momentum be defined as $\mathbf{P} = M\dot{\mathbf{R}}$ and the relative momentum $\mathbf{p} = m\dot{\mathbf{r}}$. Show that

$$\mathbf{P} = \mathbf{p}_1 + \mathbf{p}_2, \quad \mathbf{p} = \mu_1\mathbf{p}_2 - \mu_2\mathbf{p}_1 \quad \text{and} \quad \mathbf{p}_1 = \mu_1\mathbf{P} - \mathbf{p}, \quad \mathbf{p}_2 = \mu_2\mathbf{P} + \mathbf{p}. \quad (2)$$

(c) Express the total energy of the two-body central force problem (with potential $V(r)$) as a sum of center of mass and relative energies.

2. **⟨3 + 2 + 5⟩** Recall that the (relative) energy in the Kepler problem may be expressed as $E = \frac{1}{2}m\dot{r}^2 + V_{\text{eff}}(r)$ where $V_{\text{eff}} = \frac{l^2}{2mr^2} - \frac{\alpha}{r}$. Here r is the magnitude of the relative vector, m the reduced mass, $\alpha = Gm_1m_2$ and l the z component of relative angular momentum. We showed that for fixed $l \neq 0$ and $E < 0$, the orbit is an ellipse $r = \rho/(1 + \epsilon \cos \phi)$ (up to a possible rotation) where ρ is the radius of the circular orbit for that l (or semilatus rectum) and ϵ the eccentricity. (a) Find ρ in terms of l, m, α and also find the minimum value of the effective potential. (b) Express the length of the semimajor axis a in terms of ρ, ϵ . (c) Express the relative energy of this elliptical orbit in terms of ρ, ϵ and also in terms of the length of the semimajor axis a .